

Online Retail Data Analysis Report

1. Introduction

This report presents an **Exploratory Data Analysis (EDA)** of an online retail transaction dataset. The primary objective is to **clean, validate, and analyze** the dataset to extract meaningful insights related to **sales trends and customer behavior**. The analysis addresses key questions, including:

- What are the overall sales trends over time?
- Are there any extreme values or anomalies that require further investigation?
- What are the potential relationships between different variables (e.g., quantity, price, country)?

The analysis is conducted using **Python, Pandas, and Matplotlib**, with the code structured into modules to ensure reusability and maintainability.

2. Data Loading and Initial Inspection

The dataset, `online_retail.csv`, was loaded into a Pandas DataFrame. The initial inspection focused on:

- Understanding the dataset structure
- Identifying missing values
- Detecting potential outliers

Code Implementation

```
import pandas as pd

# Load dataset
df = pd.read_csv('online_retail.csv')

# Basic information
print(df.shape) # Number of rows and columns
print(df.info()) # Data types and non-null counts
print(df.describe()) # Summary statistics
```

Key Observations

- The dataset contains **541,909 rows and 9 columns**.
 - The **'Description'** and **'CustomerID'** columns have missing values.
 - The **'Quantity'** and **'UnitPrice'** columns exhibit a wide range of values, suggesting potential outliers.
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3. Data Cleaning

To ensure accuracy and reliability, several data cleaning steps were performed:

3.1 Handling Missing Descriptions

Missing values in the **'Description'** column were replaced with the placeholder **"Unknown"**, preserving other valuable information.

```
df['Description'] = df['Description'].fillna('Unknown')
```

3.2 Handling Missing Customer IDs

Similarly, missing values in **'CustomerID'** were replaced with **"Unknown"**. For customer-specific analyses, these rows will be excluded or treated separately.

```
df['CustomerID'] = df['CustomerID'].fillna('Unknown')
```

3.3 Date Parsing

The **'InvoiceDate'** column contained mixed date formats. The `dateutil.parser` module was used for flexible parsing:

```
from dateutil import parser

# Function to parse dates flexibly
def parse_dates(date_str):
    try:
        return parser.parse(date_str) # Automatically detects format
    except Exception:
        return pd.NaT # Returns NaT if parsing fails

# Apply function to InvoiceDate column
df['InvoiceDate'] = df['InvoiceDate'].apply(parse_dates)

# Validate parsing
print(f"Remaining NaT values: {df['InvoiceDate'].isna().sum()}")
print(df[['InvoiceNo', 'InvoiceDate']].head(10))
print(df['InvoiceDate'].min(), df['InvoiceDate'].max())
```

3.4 Handling Negative Quantities

Negative quantities likely indicate **returns or cancellations**. These transactions were stored separately in `df_returns` for further analysis.

```
df_returns = df[df['Quantity'] < 0]
```

3.5 Removing Non-Positive Unit Prices

Transactions with **zero or negative** prices were removed to maintain data integrity.

```
df = df[df['UnitPrice'] > 0]
```

3.6 Removing Duplicates

Duplicate records were dropped to eliminate redundancy.

```
df = df.drop_duplicates()
```

4. Data Validation

4.1 Checking for Inconsistencies

The cleaned DataFrame was checked for any remaining inconsistencies.

```
inconsistent_invoices = df[(df['InvoiceDate'].isna() |  
df['InvoiceNo'].isna())]  
print(inconsistent_invoices)
```

4.2 Analyzing Numerical Columns

```
print(df.describe())
```

The updated summary statistics provided a more accurate representation of the data after cleaning.

4.3 Identifying Extreme Values

Extreme values in 'Quantity' and 'UnitPrice' were identified and examined:

```
extreme_quantities = df[(df['Quantity'] < -1000) | (df['Quantity'] > 1000)]  
print(extreme_quantities[['InvoiceNo', 'Quantity',  
'Description']].head(10))  
  
extreme_prices = df[(df['UnitPrice'] > 1000) | (df['UnitPrice'] < 0.1)]  
print(extreme_prices[['InvoiceNo', 'UnitPrice', 'Description']].head(10))
```

Some extreme values correspond to **Amazon fees or manual entries**, suggesting they may be valid. Further investigation with domain knowledge is needed.

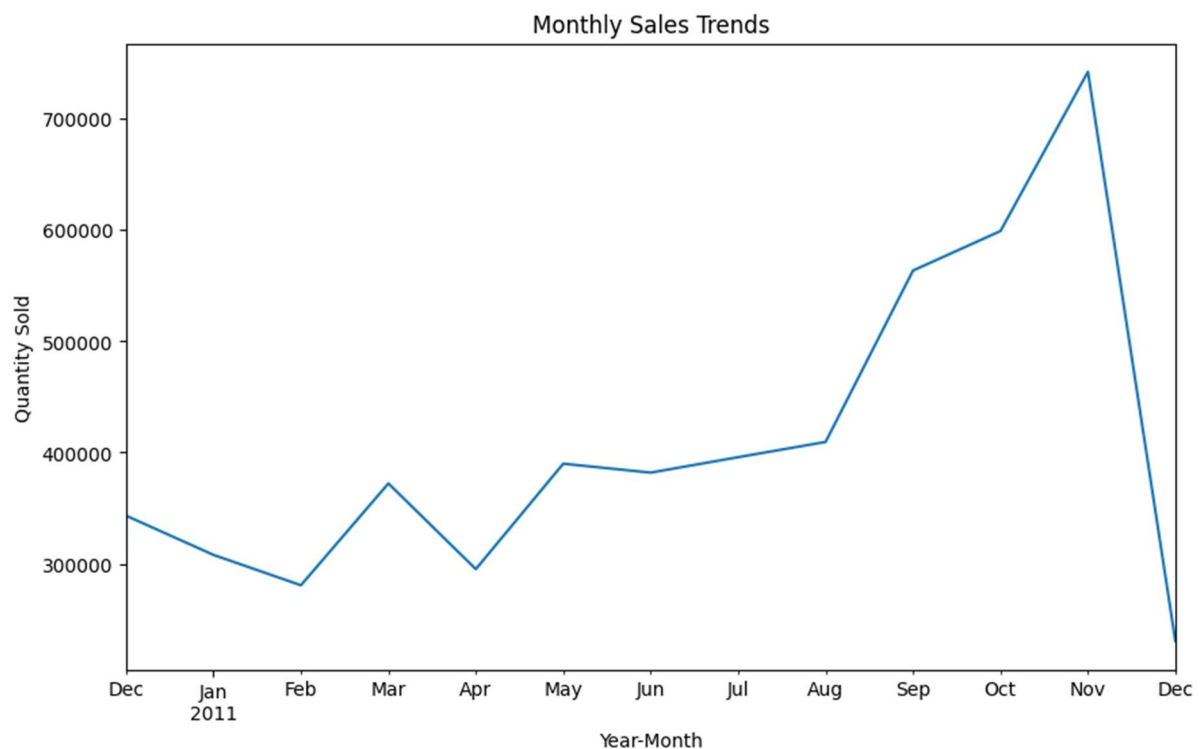
5. Data Analysis and Visualization

5.1 Monthly Sales Trends

A line plot was generated to visualize the total quantity of products sold each month throughout the dataset. This chart clearly demonstrates the seasonality in sales—with significant peaks toward the end of the year (primarily November and December) and noticeable dips in the early months (January and February). In addition, there is a subtle upward trend in the peak sales from one year to the next.

Observations:

- Strong seasonal patterns with peak sales at the end of the year.
- Noticeable declines in sales at the start of each year.
- Gradual upward shift in peak sales over successive years.



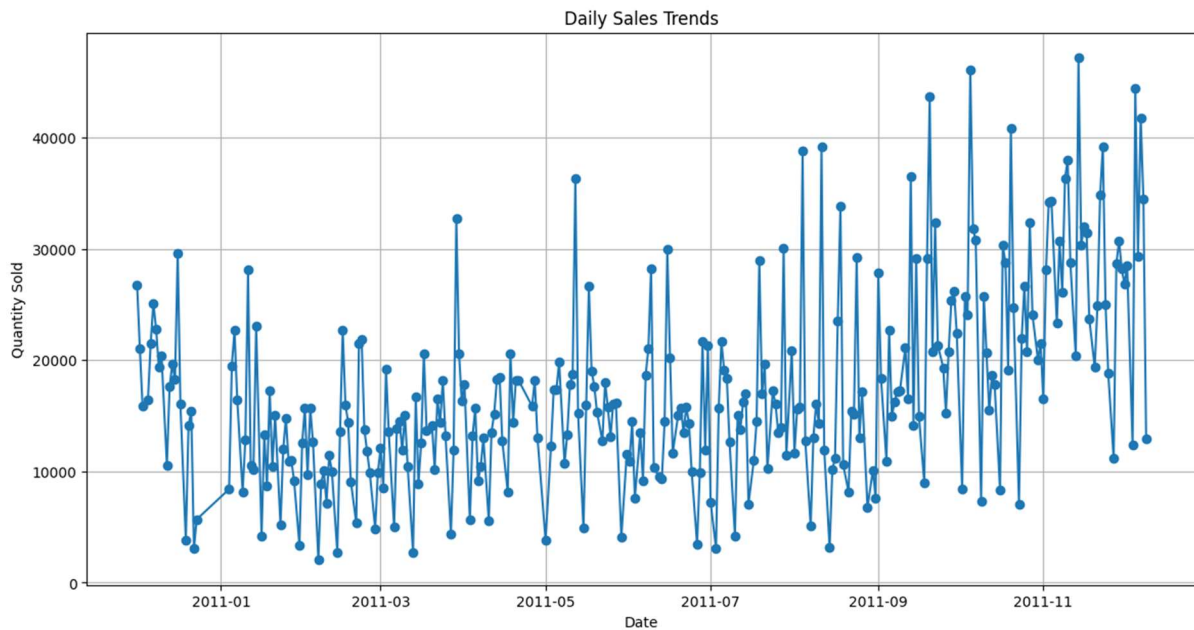
5.2 Daily Sales Trends

To obtain a more granular understanding of the sales patterns, a time series plot of daily sales was created. This visualization revealed considerable day-to-day fluctuations, confirming the seasonal trends observed in the monthly analysis. Additionally, the plot helped identify specific days or short periods with exceptionally high or low sales, which may be related to special events or anomalies.

Observations:

- Significant daily variability in sales figures.

- Confirmation of seasonal trends, with pronounced peaks near the year's end.
- Identification of outlier days that merit further investigation.

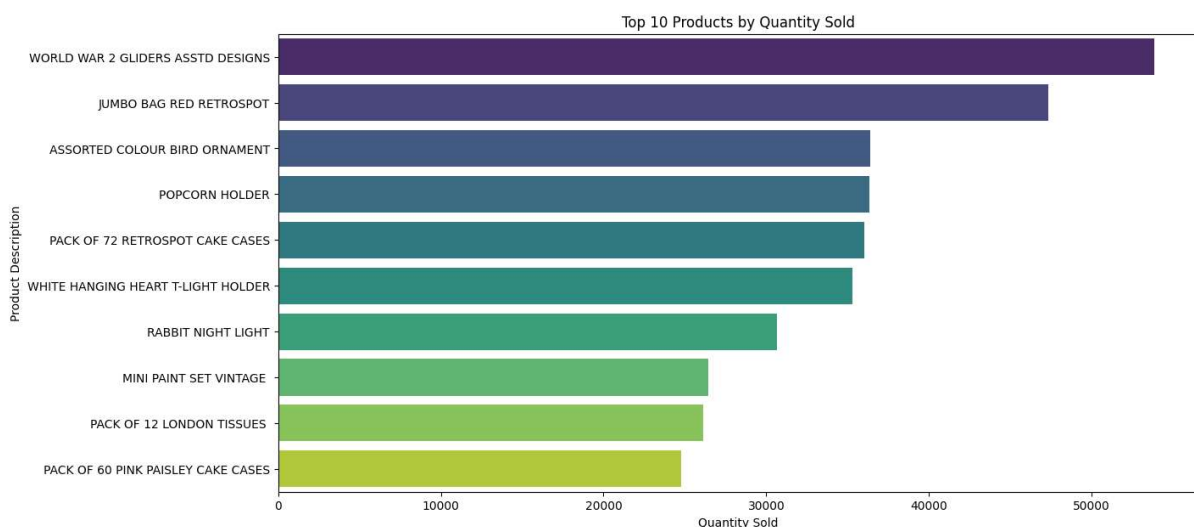


5.3 Top 10 Best-Performing Products

A bar chart was produced to highlight the top 10 products based on the total quantity sold. This analysis not only underscores which products drive overall sales but also provides insights into customer preferences. Focusing on these key products can help in refining inventory management and marketing strategies.

Observations:

- Clear identification of the products that contribute most significantly to total sales.
- Insights into customer demand patterns and product popularity.
- Potential areas to enhance product offerings or protect top performers.

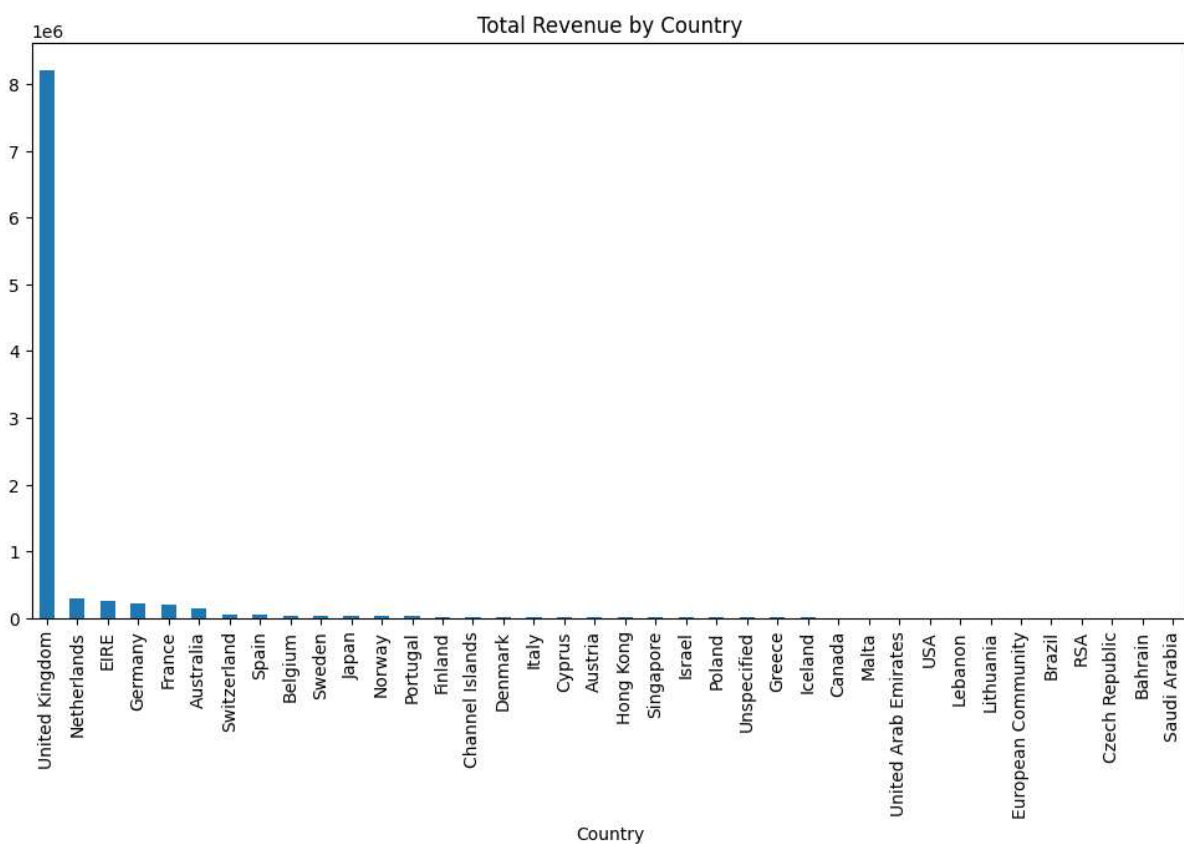


6.4 Regional Sales Analysis

A regional analysis was conducted by aggregating sales data by country. Using a bar chart (or an interactive map), this visualization exposes the disparities in sales performance across different regions. Some countries emerge as high-revenue markets, while others lag behind, indicating potential opportunities for targeted marketing or product realignment.

Observations:

- Key markets and regions are easily identifiable, UK being the main market.
- Regional consumer behavior and preferences become more apparent.
- Regions with underperforming sales may require strategic adjustments or product reconsideration.

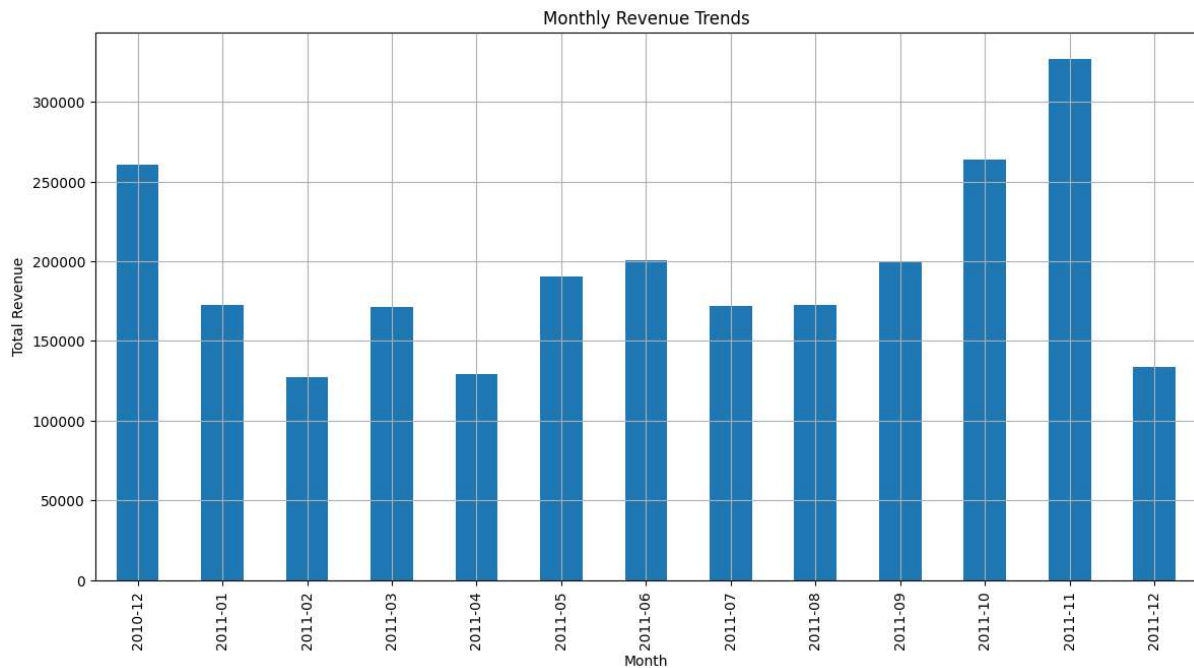


6.5 Revenue Trends by Month

The analysis also included generating a revenue trend line chart, which depicts how the total revenue evolves on a monthly basis. This visualization provides a clear overview of revenue fluctuations and helps pinpoint the months when revenue generation is at its peak.

Observations:

- Identification of high-revenue months.
- Insights that can help align marketing, sales, and product strategies with periods of peak revenue.
- Useful baseline data for forecasting and budgeting purposes.



7. Conclusion

This comprehensive analysis of the online retail dataset has yielded several valuable insights. The data exhibits seasonal sales trends, with marked peaks during the latter part of the year and troughs at the beginning. The visualizations have not only confirmed these patterns but also highlighted day-to-day variability and identified the top-performing products. Additionally, the regional sales analysis has revealed significant differences in market performance, and the revenue trend analysis has provided a clear picture of monthly revenue fluctuations. Despite some extreme values in 'Quantity' and 'UnitPrice', these outliers appear to reflect valid business scenarios (such as returns or fees) that warrant further exploration with domain-specific knowledge.

8. Next Steps

Based on the insights derived from the analysis, several future steps are recommended:

- **Customer Segmentation:**
Conduct an in-depth customer segmentation analysis to identify distinct groups based on purchasing behavior. This will enable tailored marketing efforts and product offerings.
- **Product-Specific Analysis:**
Perform a detailed study of individual product sales trends to determine the factors behind the success of top-performing items and to identify areas for improvement among underperforming products.

- **Anomaly Detection:**
Investigate the underlying causes of extreme values in 'Quantity' and 'UnitPrice' to distinguish between genuine transactions and potential data entry errors.
 - **Predictive Modeling:**
Develop forecasting models to predict future sales trends, which can inform inventory management, pricing strategies, and marketing campaigns.
 - **Geographic Analysis:**
Expand the regional analysis to uncover potential new markets for expansion, and further refine marketing strategies in both high-performing and underperforming regions.
 - **Improve Data Collection:**
Collaborate with data entry teams to enhance the accuracy of data collection.
Implementing automated validation and auto-fill scripts could reduce the need for extensive data cleaning in future datasets.
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